

Q. No. 3

$$dy/dt + 4y = 12 \quad x(0) = 2$$

$$dy/dt + ay = b$$

$$a = 4 \quad b = 12$$

Case = II

$$x(t) = Ae^{-at} + b/a$$

$$= Ae^{-4t} + 12/4$$

$$= Ae^{-4t} + 3 \quad \text{General Sol}$$

$$x(t) = [x(0) - b/a] e^{-at} + b/a$$

$$= \left[2 - \frac{12}{4} \right] e^{-4t} + 12/4$$

$$x(t) = -e^{-4t} + 3 \quad \text{Definite Sol}$$

$$-e^{-4t} = \text{complementary function } x_c$$

$$+3 = \text{particular function } x_p$$

$$\text{General Solution} = Ae^{-4t} + 3$$

$$\text{Definite Solution} = -e^{-4t} + 3$$

Q. No. 4

$$dy/dt - 5y = 0 \quad x(0) = 6$$

$$dy/dt + ay = b$$

$$a = -5 \quad b = 0$$

Case - I

$$x(t) = Ae^{-at}$$

$$= Ae^{5t} \quad \text{General Sol}$$

$$x(t) = x(0) e^{-at}$$

$$= 6 e^{5t} \quad \text{particular sol}$$